

ELECTRIC AND OPTICAL FIBRE CABLES – TEST METHODS FOR NON-METALLIC MATERIALS –

Part 402: Miscellaneous tests – Water absorption tests

1 Scope

This Part 402 of IEC 60811 describes water absorption tests which typically apply to cross-linked and thermoplastic compounds used for insulating and sheathing materials.

2 Normative references

The following documents, in whole or in part, are normatively referenced in this document and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60811-100:2012, *Electric and optical fibre cables – Test methods for non-metallic materials – Part 100: General*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in IEC 60811-100 apply.

4 Test method

4.1 General

This part of IEC 60811 shall be used in conjunction with IEC 60811-100.

Unless otherwise specified, tests shall be carried out at room temperature.

4.2 Pre-conditioning

All the tests shall be carried out not less than 16 h after the extrusion or cross-linking, if any, of the insulating or sheathing compounds.

If the test is carried out at room temperature, the test pieces shall be kept for at least 3 h at a temperature of $(23 \pm 5) ^\circ\text{C}$.

4.3 Electrical water absorption test

4.3.1 Apparatus

The following apparatus shall be used:

- a) a.c and d.c voltage sources;
- b) voltmeter;
- c) water bath with heating equipment.

4.3.2 Preparation of test pieces

The cores to be tested shall be removed from a sample of finished cable approximately 3 m long. Care shall be taken to avoid damage to the insulation during removal of the cores.

4.3.3 Test procedure

a) Pre-test

The cores shall be immersed in a water bath in which the water has been heated to the temperature specified in the standard for the type of cable.

The ends of the cores shall protrude sufficiently above the water level to prevent damage due to leakage current along the surface of the cores when the required voltage is applied between the conductors and the water.

After the cores have been immersed in the water for 1 h, an a.c. voltage of 4 kV shall be applied between the conductors and the water for 5 min. If any sample of core breaks down, it shall be removed from the water bath and not used in the main test specified in item b) below. However, the test shall be repeated not more than twice, by taking another sample of the same core, which shall be subjected to the same pre-test.

The object of the pre-test is to ensure that only undamaged cores are used for the main test.

b) Main test

Cores which are satisfactory on the pre-test shall remain in the water bath with the water still maintained at the temperature specified in the relevant standard.

A d.c. voltage in accordance with Table 1 below shall be applied between the conductors and the water for the time specified in the standard for the type of cable, the negative pole being connected to the conductor of each test piece.

Table 1 – DC test voltages

Thickness, t , of insulation specified Mean value mm	DC voltage V
0,8 and 0,9	800
1,0 and 1,2	1 000
$1,2 < t \leq 1,6$	1 400
$1,6 < t \leq 2,0$	2 000
$2,0 < t$	2 500

4.3.4 Evaluation of results

No breakdown shall occur.

4.4 Gravimetric water absorption test

4.4.1 Preparation of test piece

- a) For cables with conductors of nominal cross-sectional area equal to or less than 25 mm² and rated voltage up to and including 0,6/1 kV:

Each test piece shall be a piece of core approximately 300 mm in length.

- b) For all other cables:

Slices of 0,6 mm to 0,9 mm thickness shall be ground or cut with surfaces approximately parallel and free from roughness.

Test pieces 80 mm to 100 mm long and 4 mm to 5 mm wide shall be punched out of the slices.

- c) Two test pieces shall be prepared from each core to be tested.

4.4.2 Testing procedure

- a) For test pieces as in item a) of 4.4.1:

Clean the surface of the test piece by rubbing with a filter paper moistened with water.

Allow the test piece to dry at $(70 \pm 2)^\circ\text{C}$ to constant weight. It may also be allowed to dry by placing it for 24 h in a low-pressure oven at not more than 6,6 mbar and $(70 \pm 2)^\circ\text{C}$. Cool the sample in a desiccator.

Weigh the test piece to within 0,1 mg. Let M1 be the mass in milligrams.

Wind the test piece around a mandrel, whose diameter is between six and eight times that of the test piece, so as to bend it to a U shape and force the ends through apertures bored in the cover of a suitable glass vessel. Only the two test pieces of the same core should be in the glass vessel.

Adjust the position of the test piece such that 250 mm of its length is immersed when the vessel is filled with water up to the edge of the fitted cover.

Use preboiled distilled or deionized water.

Allow the test piece to remain at the temperature and for the time specified in the product standard. If the time is not specified, it shall be 14 days for specified thicknesses up to 1,0 mm, 21 days for thicknesses between 1,1 mm and 1,5 mm, and 28 days for thicknesses above 1,5 mm. If the temperature is not specified, it shall be 5 K below the maximum conductor temperature, but not exceeding 90°C . The water level shall be maintained up to the inside surface of the cover.

The water is then allowed to cool to ambient temperature. Remove the test piece and shake off any drops of water adhering to it, wipe lightly with a filter paper and weigh to the nearest milligram within 2 min to 3 min of its being removed from the water. Let M2 be the new mass in milligrams.

Finally, dry out the test piece under the same conditions as were used before immersion, i.e. using whichever of the two alternative methods described above had been used before the first weighing. Let M3 be the final mass in milligrams.

- b) For test pieces as in item b) of 4.4.1:

The test pieces, with thoroughly cleaned surfaces, shall be heated at $(70 \pm 2)^\circ\text{C}$ under vacuum (residual pressure close to 1 mbar) for 72 h. Materials of substantially different compositions shall not be treated in the same cell or oven at the same time.

After this treatment, the test pieces shall be cooled for 1 h in a desiccator and weighed to the nearest 0,1 mg (mass M1).

The test pieces shall then be immersed in deionized (or distilled) water at the temperature and for the time specified in the standard for the type of cable. If the temperature is not specified, it shall be 5 K below the maximum conductor temperature, but not greater than 90°C . Each test piece shall be completely immersed in a separate glass to be equipped with a condenser, or in a beaker covered with a glass lid.

If a condenser is used, its upper part shall be covered with aluminium foil to prevent any contamination.

After the time specified in the standard for the type of cable, or after 14 days if the time is not specified in the cable standard, the test pieces shall be transferred into deionized (or distilled) water at room temperature and kept there to cool. Each test piece shall then be removed from the water, shaken to detach any adherent drops, dried with special filter

paper leaving no fibres, and weighed to the nearest 0,1 mg (mass M2). Finally the test piece shall be treated under the same conditions as were used before immersion. Let M3 be the final mass in milligrams.

4.4.3 Expression of results

The mass variation in milligrams shall be calculated from one of the following formulae:

- a) if the final mass M3 is less than M1:

$$(M2 - M3)/A$$

- b) if the final mass M3 is greater than M1:

$$(M2 - M1)/A$$

where A is, for test pieces as in item a) of 4.4.1, the surface area in square centimetres of the 250 mm long immersed portion of sample, and, for test pieces as in item b) of 4.4.1, the total surface area of the immersed test piece in square centimetres.

The mean value of the mass variation of the two test pieces shall be recorded as the value for the core.

5 Test report

The test report shall be in accordance with that given in IEC 60811-100.